CAPSTONE SUBMISSION INDEX

SOLANA LOCAL EXPLORER · BITS PILANI · B.SC. COMPUTER SCIENCE 2025–2026

PROJECT	**Solana Local Explorer** — a self-hosted, one-command block explorer and real-time transaction indexer for private Solana validators
SUBMITTED BY	**Garvit Dadheech** · **Shubh Kesharwani**
SUPERVISOR	Dharma Teja
REPOSITORY	`github.com/shubhiscoding/Solana-Local-Indexer-Explorer`
DEMONSTRATION	`loom.com/share/67ac946f6bf04277842b41423a0b5730`
DATE	28 August 2026

Deliverables in this folder

FILE	TYPE	CONTENTS
`00_Submission_Index.pdf`	PDF · 3 pp	This document — what each deliverable is and where each requirement is answered.
`01_Project_Report.pdf`	PDF · 61 pp	**Final documentation.** Cover, **certificate page for supervisor signature**, declaration, acknowledgement, abstract, table of contents, list of figures and tables, nine chapters (introduction, literature review, requirements, design, implementation, **testing and validation**, results, challenges, conclusion), references and four appendices. Bookmarked and page-numbered.
`02_Project_Summary.pdf`	PDF · 9 pp	**Pre-viva briefing for the supervisor.** Project at a glance, the problem, the **unique value proposition**, the solution and architecture, key features, stack, validation summary, results and limitations, and a step-by-step demonstration guide with the questions the authors expect.
`03_Code_Submission/Solana-Local-Indexer-Explorer-source.zip`	ZIP · 1.0 MB	**Code submission.** 136 files — all source, Dockerfiles, Prisma schema and migrations, both test suites and the project documentation. Excludes `node_modules`, build output, git history and OS metadata.
`04_Source_Code_Link.txt`	TXT	**Source code link.** GitHub URL, clone and run commands, instructions for granting read access to the internal evaluator and viva supervisor if the repository is private, and the demonstration video link.
`05_User_Manual.pdf`	PDF · 20 pp	**User manual.** What the tool does and does not do, prerequisites, starting the explorer, a screen-by-screen tour, the REST API with worked examples, common tasks, configuration, troubleshooting, FAQ and a glossary.

FILE	TYPE	CONTENTS
06_Installation_Guide.pdf	PDF · 13 pp	**Installation guide.** Clean install from zero on macOS, Linux and Windows (WSL2), with five independent verification checkpoints, optional local development setup, reset and uninstall, and installation-specific troubleshooting.
07_Test_Cases_and_Validation_Report.pdf	PDF · 16 pp	**Test cases and validation report.** All 92 test cases across six levels, the executed automated results, the defect log, the requirements traceability matrix and the verbatim test console output. Also reproduced as Chapter 6 and Appendix D of the project report.
08_Final_Presentation.pptx	PPTX · 14 slides	**Final presentation** in the BITS standard template — the institutional slide master, layouts and title-slide artwork are reused unchanged. Includes speaker notes on every slide.
_source/	Folder	Reproducible build sources for these documents: the HTML/CSS the PDFs are rendered from, the SVG figures, and the Python scripts that render, page-number, bookmark and QA them. Not part of the assessed submission; included so every artefact can be regenerated.

Where each submission requirement is answered

#	REQUIREMENT	DELIVERABLE	DONE
1	Project report (final documentation), **signed by the supervisor**	01_Project_Report.pdf — the signature block is on page 2, “Certificate of Approval”, with lines for the supervisor and the viva examiner	✓
2	Summary of the project in a few pages for the viva supervisor, **UVP mandatory**	02_Project_Summary.pdf — the UVP appears on the cover and again as its own section (Section 2), with the differentiators and the competitive positioning	✓
3	Code submission (zip)	03_Code_Submission/Solana-Local-Indexer-Explorer-source.zip	✓
4	Source code link (GitHub) in a txt file, with access instructions if private	04_Source_Code_Link.txt	✓
5	Test cases and validation reports **in the final documentation**	Chapter 6 and Appendix D of 01_Project_Report.pdf; also standalone as 07_Test_Cases_and_Validation_Report.pdf and as TEST_CASES_AND_VALIDATION.md in the repository	✓
6	User manual (readme document for the project)	05_User_Manual.pdf; the repository README.md is the short form	✓
7	Installation guide	06_Installation_Guide.pdf; the repository INSTALLATION_GUIDE.md is the short form	✓
8	Final presentation in the BITS standard template	08_Final_Presentation.pptx — built on the supplied final_capstone.pptx master and layouts	✓

TWO THINGS TO DO BEFORE SUBMITTING

1. Get the certificate page signed. Print page 2 of `01_Project_Report.pdf`, have the supervisor sign and date it, and either scan it back into the PDF or submit the signed page alongside it. The viva cannot be conducted without it.

2. Grant repository access if the repository is private. `04_Source_Code_Link.txt` has two placeholders for the internal evaluator's and the viva supervisor's GitHub usernames — fill them in, or make the repository public.

VERIFICATION AN EXAMINER CAN RUN IN UNDER A MINUTE

```
cd explorer && pnpm install && pnpm test     # expect:  9 passed (9)
cd ../indexer && pnpm install && pnpm test   # expect: 29 passed (29)
```